Project Workflow

FitBuddy – AI Powered Fitness Planner

Department of Computer Science Engineering
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SmartBridge Generative AI Project

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Project Workflow – FitBuddy

1. Introduction

The project workflow defines the sequence of steps followed to design, develop, and deploy the FitBuddy application. The workflow ensures that the project is implemented in an organized and systematic manner. It covers stages such as requirement understanding, system design, development, AI integration, database management, and testing.

By following this workflow, the development team can ensure that all components of the application function smoothly and meet the intended objectives.

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2. Workflow Steps

2.1. Step 1 — Requirement Analysis

In this stage, the project requirements are identified and analyzed.

The main objective of FitBuddy is to generate personalized 7-day workout plans using AI based on user inputs such as age, weight, fitness goal, and workout intensity. Additional features such as nutrition tips and feedback-based plan updates are also defined during this stage.

2.2. Step 2 — System Design

During the system design phase, the architecture of the application is planned.

The system is divided into three main components: the frontend interface, the backend server using FastAPI, and the AI integration using the Gemini model.

SQLite is selected as the database to store user details and generated workout plans.

2.3. Step 3 — Frontend Development

In this phase, a user-friendly HTML interface is created. The frontend allows users to enter their personal details such as name, age, weight, fitness goal, and preferred workout intensity.

The interface also displays the generated workout plans and nutrition tips clearly.

2.4. Step 4 — Backend Development

The backend is developed using FastAPI. It handles user requests, processes the input data, communicates with the Gemini AI model, and returns the generated workout plans.

API endpoints are created to generate workout plans, accept feedback, and retrieve stored data.

2.5. Step 5 — AI Integration

Google Gemini AI is integrated into the backend to generate personalized workout plans and nutrition tips. The backend sends structured prompts containing user details to the AI model, which then returns a customized 7-day fitness schedule.

2.6. Step 6 — Database Integration

SQLite is used to store user profiles and their generated workout plans.

This allows users to retrieve their plans later and also enables the system to update workout routines based on user feedback.

2.7. Step 7 — Feedback Processing

Users can provide feedback on their workout plans, such as requesting more cardio exercises or additional rest days.

The backend processes this feedback and sends it to the Gemini AI model to generate an updated workout plan that better suits the user's needs.

2.8. Step 8 — Testing and Deployment

In the final stage, the application is tested to ensure that all components function correctly.

The API endpoints, database storage, and user interface are verified. After successful testing, the system is ready for deployment and user interaction.

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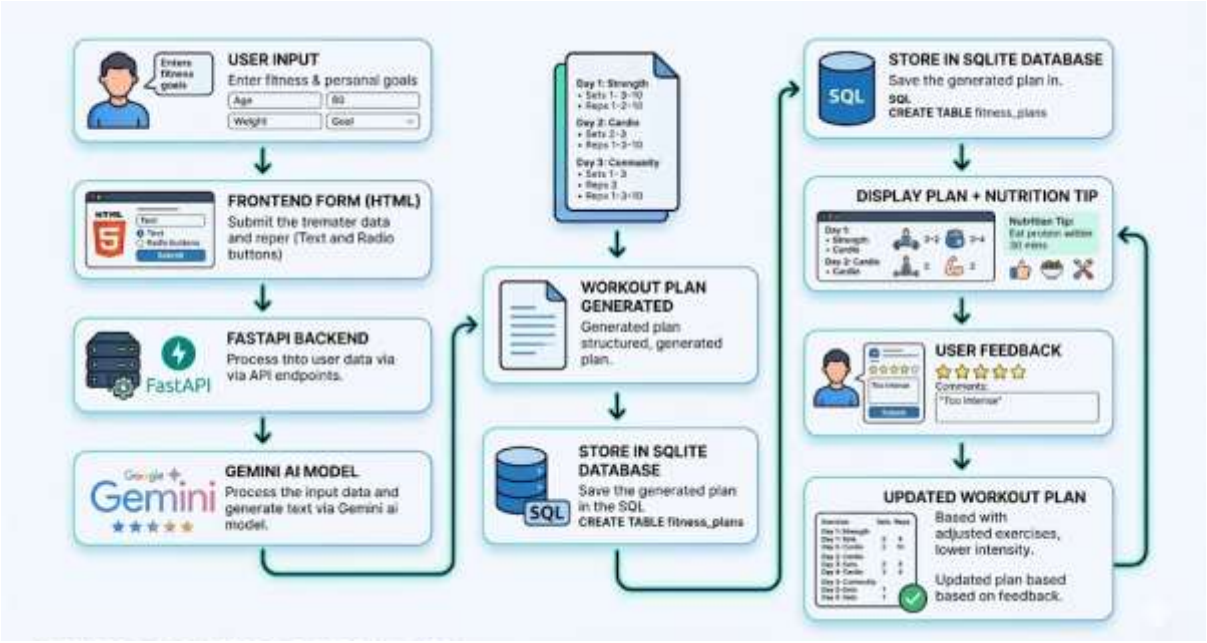
3. Workflow diagram